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Bank Bonus Pay as a Risk Sharing Contract

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1 Big picture

- **Question.** Are banker bonuses only incentive pay, or do they also help banks share risk with employees when financing conditions deteriorate?
- **Main answer.** Bonus pay has an important bankwide risk-sharing role. Because external finance is costly in bad states, shareholders value internal cash more when banks are constrained. Variable pay lets the bank reduce labor costs exactly when internal cash is scarce.
- **Theoretical idea.** Holmstrom-style incentive pay is combined with financial frictions. In unconstrained states, employees can receive high incentive-sensitive pay and shareholders receive dividends. In constrained states, dividends are suspended and the whole wage bill scales down, so employees share aggregate bank risk.
- **Data.** The paper uses payroll records from 327 Austrian, German, and Swiss commercial banks over 2003–2010. The data contain about 1.26 million employee-year observations below the executive level, covering hierarchy levels and functional divisions such as logistics, headquarters, retail banking, corporate banking, private banking, asset management, investment banking, and treasury/capital markets.
- **Why the setting is powerful.** The data separately observe fixed base salary and variable bonus pay for nonexecutive employees in all bank functions. This makes it possible to distinguish incentive-heavy front-office roles from support employees whose pay should not be tightly linked to their own measurable performance.
- **Main descriptive fact.** Bonuses are widespread even in internal service functions such as IT, accounting, and human resources. The existence of variable pay in these jobs is hard to explain purely through individual incentives.
- **Variance-decomposition result.** Incentive pay accounts for much of pay variation, especially in senior and trading jobs, but risk sharing accounts for a sizable share. For junior employees and support/retail divisions, risk sharing is quantitatively important.
- **Crisis result.** Employees with higher precrisis bonus shares experienced larger compensation cuts in 2008. The cuts are stronger in banks that became more financially constrained.
- **Cross-divisional result.** Employees' bonus shares respond not only to their own division's earnings but also to earnings in other divisions. Loan officers benefit from trading income; trading employees and support employees benefit from lending income; service employees' bonuses fall when loan loss provisions rise.
- **Financial-stability implication.** Bonus pay can reduce operating leverage by making labor costs state-contingent. Banning or severely capping bonuses may unintentionally make banks' labor costs more rigid and weaken a private buffer against distress.
- **Takeaway.** The paper does not deny incentive motives. It argues that the standard incentive-only view is incomplete: bank bonuses can simultaneously provide incentives and insure shareholders against bad aggregate states by sharing risk with employees.

2 Data and measurement

2.1 Payroll data

- The compensation data are directly sourced from bank payroll records collected by Willis Towers Watson.
- The sample covers 54 Austrian, 185 German, and 88 Swiss commercial banks from 2003 to 2010.
- The paper observes more than 1.26 million employee-year observations below the executive level.
- For each employee-year, the data separate total compensation into fixed base salary and annual bonus.
- Banks are required to report data for at least 80% of nonexecutive employees, and the data provider validates submissions with compensation consultants.

Interpretation: The paper’s strength is that it observes compensation at the employee level rather than only at the executive or bank level. This lets the authors test whether aggregate bank shocks affect employees who are far from the income-generating desks.

2.2 Employee classifications

- Employees are classified into standardized occupations and functional divisions.
- The paper groups functions into internal services, loan business, wealth management, and trading business.
- Employees are also assigned to six hierarchy levels based on skills, responsibility, and position in the career ladder.
- Age and tenure are observed and used to form labor-market controls.

Interpretation: Division, hierarchy, age, and tenure controls are crucial because the empirical question is not whether traders earn more than back-office workers. The question is whether two employees with similar labor-market characteristics experience different bonus cuts because their banks are financially constrained.

2.3 Compensation variables

The main compensation objects are

$$w_{k,t} = \text{annual total compensation of employee } k, \quad (1)$$

$$\bar{w}_{k,t} = \text{fixed base salary}, \quad (2)$$

$$Bonus_{k,t} = w_{k,t} - \bar{w}_{k,t}, \quad (3)$$

$$BonusShare_{k,t} = \frac{Bonus_{k,t}}{\bar{w}_{k,t}}. \quad (4)$$

Interpretation: The bonus share is more informative than the raw bonus because it measures how much of an employee’s compensation is flexible relative to the fixed wage claim. A high bonus share means the employee has fewer debt-like fixed claims on the bank.

2.4 Bank accounting data

The payroll data are matched to bank-level accounting data from Bankscope. Key bank-level variables include:

- assets and log assets;

- loans/assets, deposits/assets, and equity/assets;
- lending income per employee;
- trading income per employee;
- loan loss provisions per employee;
- short-term debt funding, off-balance-sheet activity, and cash changes around the crisis.

Interpretation: The accounting data identify the bankwide resources available to pay employees. In the risk-sharing view, bonuses should depend on bankwide resource shocks even if those shocks originate outside an employee's own division.

2.5 Employee panel

- A smaller subsample tracks individual employees over time from 2003 to 2007.
- This panel is available mainly for German banks and ends before the crisis.
- It allows specifications with employee fixed effects to control for persistent employee productivity and innate ability.

Interpretation: The employee panel is used to strengthen the cross-divisional tests outside the crisis. It cannot be used for the 2008 crisis design because the identifiers are not available in the crisis years.

3 Model and empirical specifications

3.1 Basic risk-sharing mechanism

The paper extends an incentive-pay model by adding bank refinancing frictions. In good states, the bank can pay employees according to their unconstrained incentive contracts and distribute residual resources to shareholders. In bad states, dividends cannot be negative and outside finance is costly, so the wage bill must absorb part of the shock.

A simple way to write the decomposition is

$$w_k(s, x) = \Lambda_k(s, e_k) w(s), \quad (5)$$

where $\Lambda_k(s, e_k)$ is the employee-specific incentive component and $w(s)$ is the bankwide wage scale.

Taking logs gives the key decomposition:

$$\ln w_k(s, x) = \ln \Lambda_k(s, e_k) + \ln w(s). \quad (6)$$

Interpretation: The first term is the incentive component: employees with high marginal contributions or high incentive needs receive higher relative pay. The second term is common to all employees and represents risk sharing through the bankwide wage bill.

3.2 Variance decomposition

Conditioning on labor-market state C , the paper decomposes log wage variance as

$$\text{Var}[\ln w_k|C] = \text{Var}[\ln \Lambda_k|C] + \text{Var}[\ln w|C] + 2\text{Cov}[\ln \Lambda_k|C, \ln w|C]. \quad (7)$$

The percentage contribution of the risk-sharing component is

$$\frac{\text{Var}[\ln w|C]}{\text{Var}[\ln w_k|C]}. \quad (8)$$

Interpretation: If a large share of pay variation is explained by the bankwide wage component, then bonuses are not merely individual performance pay. They also adjust compensation to bankwide resources.

3.3 Base salary, bonus, and bonus share

The model defines base salary as the lower bound of employee pay across possible realizations of performance. Bonus pay and bonus share are

$$\text{Bonus}_k(s) \equiv w_k(s, x) - \bar{w}_k(s), \quad (9)$$

$$\text{BonusShare}_k(s) \equiv \frac{\text{Bonus}_k(s)}{\bar{w}_k(s)} = \frac{w_k(s, x) - \bar{w}_k(s)}{\bar{w}_k(s)}. \quad (10)$$

Interpretation: Base salary is the rigid part of labor cost. The bonus share is the flexible part of labor cost. A bank with high bonus shares has lower operating leverage because compensation can fall when financial resources fall.

3.4 Crisis labor-cost regression

The 2008 crisis test uses employee-group observations. Groups are defined by bank, division, hierarchy level, age group, and tenure group. A generic specification is

$$\Delta w_{g,2008} = \beta \text{BonusShare}_{g,2007} + \theta (\text{BonusShare}_{g,2007} \times \text{Constraint}_{b,2008}) + \Gamma' X_{b,2007} + \mu_{d \times h \times a \times ten} + \alpha_b + \varepsilon_g, \quad (11)$$

where

$$\Delta w_{g,2008} = \frac{w_{g,2008} - w_{g,2007}}{w_{g,2007}}. \quad (12)$$

Interpretation: The coefficient β asks whether employees with more flexible compensation experience larger pay cuts. The interaction θ asks whether this effect is stronger in banks that are more financially constrained.

3.5 Cross-divisional risk-sharing regression

The core cross-divisional specification is

$$\text{BonusShare}_{k,b,t} = \beta_L \frac{\text{LendingIncome}_{b,t}}{\text{Employees}_{b,t}} + \beta_T \frac{\text{TradingIncome}_{b,t}}{\text{Employees}_{b,t}} + \Gamma' X_{b,t} + FE + \varepsilon_{k,b,t}. \quad (13)$$

Interpretation: The test is strongest when employees outside the income-generating division respond to earnings. If internal service employees receive higher bonuses when lending or trading income rises, the bonus pool is operating at the bankwide level.

3.6 Loan-loss provision regression

The precrisis employee-panel test also estimates

$$\text{BonusShare}_{k,b,t} = \beta_{LLP} \frac{LLP_{b,t}}{\text{Employees}_{b,t}} + FE + \varepsilon_{k,b,t}. \quad (14)$$

Interpretation: Loan loss provisions are negative shocks to lending income. A negative coefficient implies that bonuses fall when realized loan losses rise, even for employees outside lending.

3.7 Testable predictions

1. **Sources of pay variation:** employee pay varies for both incentive and risk-sharing reasons.
2. **Labor-cost reduction in crisis:** employees with higher precrisis bonus shares experience larger compensation cuts in 2008; the effect is stronger in constrained banks.
3. **Risk sharing across divisions:** bonus shares depend on bankwide lending and trading income, not only on own-division income.
4. **Bankwide crisis sharing:** even employees with little responsibility for the crisis exposure, such as service employees and junior employees, experience bonus cuts during the crisis.

Interpretation: The predictions distinguish risk sharing from a pure incentive story. Incentive pay predicts performance sensitivity for employees who affect output; risk sharing predicts pay sensitivity to bankwide shocks even for employees who do not directly create those shocks.

4 Identification and empirical design

4.1 What identifies the risk-sharing channel

- The main evidence combines employee-level pay data, bank-level accounting shocks, and division-level job classifications.
- The empirical tests ask whether employees' bonuses respond to bankwide shocks that are outside their individual control.
- Labor-market controls compare employees in similar divisions, hierarchy levels, ages, and tenures.
- The employee-panel tests add employee fixed effects, which absorb time-invariant worker quality.

Interpretation: The design is not randomized, but it is built around a sharp distinction: individual incentives should not make an IT worker's bonus respond strongly to trading income, whereas bankwide risk sharing can.

4.2 Separating risk sharing from labor-market wage flexibility

- A pure labor-market channel would say that bonuses fall in 2008 because outside options for all bank employees deteriorate.
- The paper addresses this by using division $\times$ hierarchy $\times$ age $\times$ tenure fixed effects.
- It also shows that service employees outside finance did not experience the same wage contraction, making a general labor-market explanation less plausible.
- Crisis regressions interact precrisis bonus share with bank-specific financial constraint measures.

Interpretation: The strongest crisis evidence is not simply that bank pay fell in 2008. It is that more bonus-exposed workers at more constrained banks experienced larger cuts.

4.3 Financial constraint measures

The crisis tests proxy for constraints using changes from 2007 to 2008 in

- short-term debt funding;
- off-balance-sheet activity;
- cash.

Interpretation: These variables are not perfect measures of constraints, but they capture funding pressure and liquidity needs around a crisis when recapitalization was costly.

4.4 Threats to interpretation

- **Employee productivity:** more productive workers may be sorted into banks that later become constrained. Table 8 addresses this by comparing precrisis compensation across constrained and unconstrained banks.
- **Persistent worker heterogeneity:** employee fixed effects in Tables 6 and 7 address time-invariant worker quality.
- **Outside options:** labor-market fixed effects and the nonfinancial service-worker comparison reduce concern that all results are driven by reservation wages.

- **Incentive pay overlap:** the paper does not claim bonuses are never incentives. The evidence is that some bonus sensitivity cannot be explained by incentives alone.

Interpretation: The paper's credibility comes from the accumulation of patterns: variance decomposition, crisis cuts, cross-division spillovers, loan-loss sensitivity, and service-worker comparisons all point toward risk sharing.

5 Tables and figures with detailed interpretations

This section groups adjacent visuals where possible. All images are cropped directly from the original paper.

5.1 Figure 1: model intuition

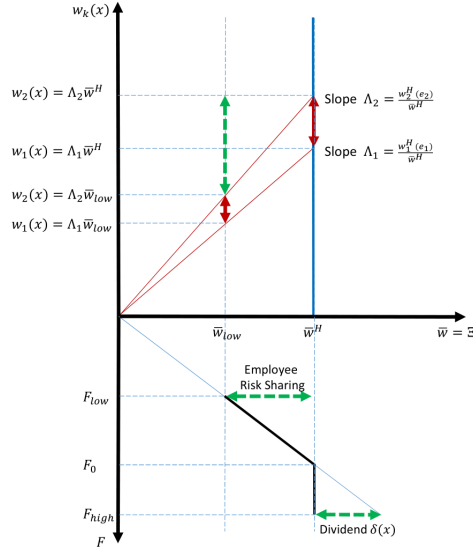


Figure 1: Risk sharing in the model.

Read:

- The vertical axis is employee-specific compensation; the horizontal axis is the bankwide average wage.
- Employees have different relative Holmstrom compensation slopes, so higher-powered employees have larger pay movements.
- When bank resources are below the threshold F_0 , dividends are suspended and employee pay scales down.
- When resources exceed F_0 , employees reach capped compensation and additional resources are paid as dividends.

Economic message: The figure is the paper's central economic mechanism. Bonus pay is not only an incentive device; it is a contingent claim that lets employees absorb part of bad bankwide shocks. The more compensation is bonus-like, the more the employee contributes to bank-level risk sharing.

Interpretation: The figure also clarifies why the policy debate is subtle. Capping variable pay may reduce excessive risk-taking incentives, but it can also make labor costs more fixed and increase bank operating leverage in distress.

5.2 Table 1 and Table 2: descriptive pay structure and pay determinants

Table 1

Summary statistics

A. Employee-year statistics in the full sample (2003–2010)

	Obs.	Mean	SD	P10	P25	P50	P75	P90
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
w (Total compensation)	1,261,693	70,095	42,397	38,766	46,367	59,196	79,624	110,849
$\bar{w}$ (Base salary)	1,262,994	60,014	23,639	37,068	44,092	54,648	69,882	89,195
Bonus	1,269,208	10,138	24,785	0	942	3,873	9,522	23,478
Bonus share	1,262,145	0.129	0.209	0.000	0.020	0.077	0.152	0.302
Tenure	1,196,031	13	10	2	5	11	19	28
Age	1,096,976	41	9	28	34	41	48	54

B. Employee-year statistics in the employee panel (2003–2007)

	Obs.	Mean	SD	P10	P25	P50	P75	P90
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
w (Total compensation)	124,632	64,023	44,042	36,755	42,892	53,263	72,600	97,375
$\bar{w}$ (Base salary)	124,632	53,451	19,547	34,905	40,820	49,261	61,490	76,700
Bonus	124,632	10,688	30,589	0	968	4,840	10,563	22,000
Bonus share	124,632	0.151	0.254	0.000	0.024	0.108	0.179	0.323
Tenure	103,745	15	10	4	7	15	21	30
Age	114,817	41	9	29	35	41	48	53

C. Bonus share and base salary by bank division in the full sample (2003–2010)

	Bonus share				Base salary			
	Obs.	Mean	SD	P50	Obs.	Mean	SD	P50
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Internal services								
D1 Logistical services	288,580	0.068	0.078	0.051	288,591	49,365	15,154	46,500
D2 HQ services	371,715	0.133	0.145	0.091	371,970	67,631	21,834	63,600
Loan business								
D3 retail banking	374,658	0.083	0.094	0.066	374,820	49,590	14,581	46,980
D4 corp. banking	58,469	0.182	0.188	0.127	58,462	72,183	24,640	67,340
Wealth management								
D5 private banking	78,356	0.265	0.290	0.169	78,361	78,615	28,727	73,332
D6 asset mgmt.	22,112	0.366	0.377	0.263	22,520	84,863	30,449	80,000
Trading business								
D7 invest. banking	13,086	0.420	0.614	0.212	13,086	89,696	37,894	80,042
D8 Treasury/markets	37,004	0.427	0.650	0.191	37,003	86,068	36,859	77,700
Junior employees	742,758	0.062	0.076	0.049	742,943	47,489	11,676	46,345
Senior employees	519,387	0.225	0.287	0.152	520,051	77,906	24,863	71,980

Table 1: Table 1, panels A–C.

Table 2
Determinants of bank employee pay

Dependent variable:	log ω			Bonus share		
	(1)	(2)	(3)	(4)	(5)	(6)
Age / 10	0.354*** (0.015)		0.357*** (0.018)	0.058*** (0.010)		0.031*** (0.006)
(Age / 10) ²	−0.035*** (0.002)		−0.036*** (0.002)	−0.007*** (0.001)		−0.005*** (0.001)
Tenure / 10		0.072*** (0.006)	0.003 (0.005)		0.035*** (0.009)	0.034*** (0.010)
(Tenure / 10) ²		−0.009*** (0.001)	0.002 (0.001)		−0.009*** (0.002)	−0.008*** (0.002)
Hierarchy FE relative to level 1:						
Hierarchy 2	0.137*** (0.010)	0.150*** (0.010)	0.135*** (0.009)	0.008 (0.006)	0.009* (0.006)	0.007 (0.005)
Hierarchy 3	0.312*** (0.011)	0.343*** (0.011)	0.310*** (0.010)	0.023*** (0.008)	0.025*** (0.008)	0.022*** (0.008)
Hierarchy 4	0.532*** (0.014)	0.579*** (0.015)	0.530*** (0.014)	0.079*** (0.014)	0.081*** (0.014)	0.079*** (0.014)
Hierarchy 5	0.822*** (0.020)	0.879*** (0.023)	0.820*** (0.020)	0.200*** (0.028)	0.200*** (0.027)	0.199*** (0.028)
Hierarchy 6	1.217*** (0.021)	1.280*** (0.023)	1.216*** (0.021)	0.400*** (0.044)	0.398*** (0.042)	0.399*** (0.044)
Bank division FE relative to logistical services:						
HQ services	0.073*** (0.007)	0.068*** (0.008)	0.075*** (0.008)	0.002 (0.005)	0.005 (0.004)	0.004 (0.005)
Retail banking	0.032*** (0.011)	0.011 (0.012)	0.031*** (0.011)	0.002 (0.005)	0.004 (0.006)	0.002 (0.006)
Corporate banking	0.115*** (0.012)	0.102*** (0.013)	0.116*** (0.012)	0.022*** (0.008)	0.024*** (0.007)	0.023*** (0.007)
Private banking	0.153*** (0.020)	0.140*** (0.020)	0.154*** (0.020)	0.095*** (0.030)	0.099*** (0.031)	0.097*** (0.030)
Asset management	0.272*** (0.020)	0.266*** (0.021)	0.274*** (0.020)	0.155*** (0.026)	0.162*** (0.026)	0.160*** (0.026)
Investment banking	0.336*** (0.033)	0.326*** (0.033)	0.339*** (0.033)	0.196*** (0.043)	0.203*** (0.043)	0.201*** (0.043)
Treasury/markets	0.365*** (0.021)	0.352*** (0.021)	0.367*** (0.021)	0.257*** (0.035)	0.262*** (0.036)	0.260*** (0.035)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Bank FEs	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	1,045,077	1,045,077	1,045,077	1,045,416	1,045,416	1,045,416
No. of banks	248	248	248	248	248	248
Adjusted R^2	.81	.81	.79	.42	.42	.42

Table 2: Table 2.

Table 1

(Continued)

D. Bank-year statistics (2003–2010)

	Obs.	Mean	SD	P10	P25	P50	P75	P90
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Assets (in EUR million)	273	171,465	378,978	2,891	8,251	40,165	158,889	357,657
log assets	273	10.504	1.897	7.969	9.018	10.601	11.976	12.787
Deposits/Assets	273	0.706	0.175	0.449	0.605	0.715	0.851	0.911
Loans/Assets	273	0.393	0.234	0.073	0.206	0.371	0.568	0.716
Equity/Assets	273	0.055	0.038	0.019	0.029	0.042	0.070	0.105
Trading income/Employees	273	0.016	0.258	−0.015	0.000	0.004	0.032	0.128
Lending income/Employees	273	1.629	1.784	0.162	0.339	0.857	2.355	4.584
(Lending+Trading income)/Employees	273	1.703	1.947	0.174	0.359	0.847	2.268	4.585
ΔLLP /Employees	273	0.027	0.312	−0.085	−0.017	0.000	0.024	0.114

Table 3: Table 1, panel D: bank-year statistics.

Read:

- Table 1 shows an average total compensation of about EUR 70,095 and an average bonus share of 12.9% in the full employee-year sample.
- Bonus shares are not confined to trading or investment banking. Logistical services have a mean bonus share of 6.8%, headquarters services 13.3%, and retail banking 8.3%.
- Bonus shares rise sharply with seniority: junior employees average 6.2%, while senior employees average 22.5%.
- Trading-oriented divisions have the highest bonus shares, with investment banking at 42.0% and treasury/capital markets at 42.7%.
- Table 2 confirms in regressions that hierarchy and division strongly predict both log pay and bonus shares.

Economic message: The descriptive evidence already suggests a dual role for bonuses. High trading bonuses are consistent with incentives, but positive bonuses in support functions are harder to justify as individual performance pay and point toward bankwide bonus-pool risk sharing.

Interpretation: Table 2 is important because it shows the data provider’s employee classifications are informative. Hierarchy and division explain a large share of compensation variation, which makes later labor-market controls more credible.

5.3 Table 3: variance decomposition of pay

Table 3
Variance decomposition of employee pay

Observations		(log) Pay variance		Percentage variance contribution			Ratio
Number	%	$Var[\ln w_{i,t} C]$	(2) × (3)	Incentive component $Var[\ln \Delta_{i,t} C]$	Risk sharing component $Var[\ln \Delta_{i,t} C]$	Covariance component $2Cov[\ln \Delta_{i,t} C]$	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Employee (log) total pay variance decomposition by hierarchy level							
Hierarchy 1 (low)	9%	0.022	0.002	0.667	0.442	-0.109	0.663
Hierarchy 2	25%	0.025	0.006	0.729	0.348	-0.077	0.478
Hierarchy 3	25%	0.034	0.009	0.741	0.291	-0.032	0.393
Hierarchy 4	25%	0.047	0.012	0.784	0.212	0.004	0.270
Hierarchy 5	12%	0.077	0.009	0.814	0.142	0.044	0.174
Hierarchy 6 (high)	4%	0.090	0.003	0.872	0.126	0.002	0.145
All	100%	0.041	0.076	0.776	0.239	-0.015	0.308
B. Employee (log) total pay variance decomposition by bank division							
Logistical services	23%	0.035	0.008	0.664	0.342	-0.006	0.514
HQ services	30%	0.043	0.013	0.727	0.245	0.028	0.338
Retail banking	30%	0.022	0.007	0.808	0.262	-0.070	0.325
Corp. banking	5%	0.047	0.002	0.809	0.185	0.026	0.204
Private banking	6%	0.062	0.004	0.807	0.185	0.008	0.320
Asset mgmt.	2%	0.093	0.002	0.850	0.192	-0.042	0.226
Invest. banking	1%	0.151	0.003	0.996	0.093	0.011	0.104
Treasury/markets	3%	0.155	0.005	0.879	0.076	0.045	0.086
All	100%	0.041	0.076	0.776	0.239	-0.015	0.308

We decompose the part of (log) pay variation not explained by a set C of labor market proxies into its incentive pay and its risk sharing component. To this end, we first correct total annual compensation $w_{i,t}$ for variation explained by $C = Age \times Tenure \times Division \times Hierarchy \times Year FE$ and then calculate the variance of (log) residual pay $Var[\ln w_{i,t} | C]$ reported in column 3. Columns 5, 6, and 7 report the percentage variance contribution of the incentive component $Var[\ln \Delta_{i,t} | C]$, the risk sharing component $Var[\ln \Delta_{i,t} | C]$, and the covariance component $2Cov[\ln \Delta_{i,t} | C]$. Column 8 shows the percentage variance contribution of each hierarchy level (panel A) and each bank division (panel B) to the overall variance calculated across all bank employees. Column 8 reports the variance ratio of the risk sharing component relative to the incentive component. Internet Appendix A describes the details of these calculations.

Table 4: Variance decomposition of employee pay.

Read:

- Table 3 decomposes log total pay variation into incentive, risk-sharing, and covariance components.
- At low hierarchy levels, risk sharing contributes a large fraction of pay variation. For hierarchy level 1, the risk-sharing component is reported at 44.2%.
- For hierarchy levels 2 and 3, risk-sharing contributions are also economically meaningful, around 34.8% and 29.1%.
- Higher hierarchy levels have larger incentive components, consistent with incentive pay becoming more important as responsibility rises.
- By division, risk sharing is especially relevant in internal services and retail banking, while trading has a much larger incentive component relative to risk sharing.

Economic message: This table is the quantitative bridge between the theory and the data. It shows that the bankwide wage component is not a minor residual; for many employees it explains a meaningful share of pay risk.

Interpretation: The results reconcile two views. The incentive view is right for senior and trading employees, but incomplete for the median employee. For support and retail employees, a large part of variable pay looks more like exposure to bankwide resources than direct incentive pay.

5.4 Figure 2, Figure 3, and Table 4: operating leverage in the crisis

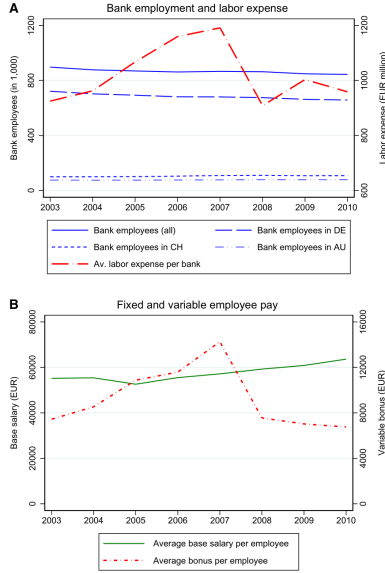


Figure 2: Figure 2.

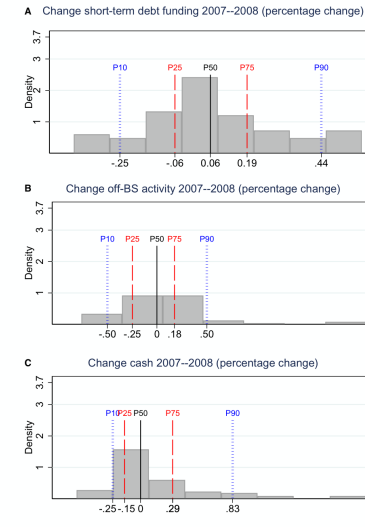


Figure 3: Bank heterogeneity in the crisis
Panels A to C show histograms for three bank variables that capture the crisis shock to banks' financial (funding) constraints. Change *Short-term Debt Funding* measures the percentage change in short-term debt funding (deposits and wholesale debt) from 2007 to 2008. Change *Off-BS Activity* the percentage change in off-balance

Table 4
Precrisis bonus pay, financial constraints, and labor cost reductions in the crisis

Dependent variable:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Bonus share</i> ₂₀₀₇	-0.264*** (0.029)	-0.265*** (0.032)	-0.271*** (0.031)	-0.280*** (0.027)	-0.273*** (0.032)	-0.273*** (0.029)
Δ <i>ST debt funding</i> 2008			0.005 (0.063)			
<i>Bonus share</i> ₂₀₀₇ × Δ <i>ST debt funding</i> 2008			0.143*** (0.033)	0.239*** (0.044)		
Δ <i>Off-BS activity</i> 2008					0.071*** (0.025)	
<i>Bonus share</i> ₂₀₀₇ × Δ <i>Cash</i> 2008						0.106*** (0.019)
Bank characteristics in 2007:						
<i>Leases / Assets</i>		-0.052 (0.046)	-0.046 (0.050)			
<i>Deposits / Assets</i>		0.104 (0.074)	0.098 (0.070)			
<i>Equity/Assets</i>		0.275 (0.360)	0.139 (0.313)			
<i>Log assets</i>		-0.003 (0.007)	-0.001 (0.006)			
Division × Hierarchy × Age	Yes	Yes	Yes	Yes	Yes	Yes
Country FEs	Yes	Yes	Yes	No	No	No
Bank FEs	No	No	No	Yes	Yes	Yes
No. of observations	7,218	6,127	6,127	6,127	5,639	6,127
No. of banks	66	44	44	44	40	44
Adjusted R ²	.27	.27	.28	.39	.38	.38

Table 5: Table 4.

Read:

- Figure 2 shows that average bonuses fell sharply in the financial crisis while base salaries were much more stable.
- Figure 3 shows substantial cross-bank variation in changes in short-term debt funding, off-balance-sheet activity, and cash from 2007 to 2008.
- Table 4 shows that higher precrisis bonus share predicts larger 2008 compensation cuts: the main coefficient is around -0.26 to -0.28 .
- Interactions with financial-constraint measures are positive and significant. Because several constraint variables are coded as changes, the economic interpretation is that bonus-exposed employees are especially affected when bank resources or funding conditions worsen.
- A one-standard-deviation higher precrisis bonus share is interpreted in the paper as leading to a roughly 10% stronger compensation reduction from 2007 to 2008.

Economic message: The crisis evidence is the operating-leverage result. Bonuses made labor costs flexible when banks faced funding stress. Fixed salaries acted like debt claims; bonuses acted like state-contingent claims.

Interpretation: The crisis test matters because it links compensation design to bank resilience. If banks had relied more on fixed salary and less on bonuses, the same negative shock would have produced a more rigid wage bill and more pressure to raise external capital or cut employment.

5.5 Table 5, Table 6, and Table 7: cross-divisional risk sharing

Table 5
Risk sharing across bank divisions

Dependent variable:	Bonus share				
	All D1-D8 (1)	Internal services D1+D2 (2)	Loan business D3+D4 (3)	Trading business D7+D8 (4)	Wealth mgmt. D5+D6 (5)
Lending income/Employees	0.026*** (0.006)	0.012** (0.005)	0.032*** (0.005)	0.091 (0.088)	0.062*** (0.007)
Trading income/Employees	0.020*** (0.005)	0.013*** (0.004)	0.013** (0.005)	0.300** (0.142)	0.039*** (0.006)
Loans/Assets	0.066 (0.084)	0.042 (0.042)	0.051 (0.057)	0.316 (0.671)	0.008 (0.215)
Deposits/Assets	0.071* (0.042)	0.084** (0.039)	-0.006 (0.038)	0.514** (0.250)	0.082 (0.095)
Equity/Assets	-0.282 (0.321)	-0.053 (0.215)	-0.619* (0.365)	-2.013 (3.352)	-0.781 (0.900)
log assets	-0.028 (0.022)	-0.029 (0.020)	-0.045*** (0.011)	-0.058 (0.179)	-0.037 (0.068)
Bank $\times$ Division $\times$ Hierarchy FEs	Yes	Yes	Yes	Yes	Yes
No. of observations	906,707	463,667	352,016	36,442	54,582
No. of banks	82	82	65	71	64
Adjusted R^2	.50	.54	.61	.31	.47

Table 6: Table 5.

Table 6
Risk sharing across bank divisions outside the crisis

Dependent variable:	Bonus share				
	All D1-D8 (1)	Internal services D1+D2 (2)	Loan business D3+D4 (3)	Trading business D7+D8 (4)	Wealth mgmt. D5+D6 (5)
<i>A. Standard errors clustered by bank and by labor market</i>					
Lending income/Employees	0.078*** (0.024)	0.048 (0.028)	0.103 (0.074)	0.485*** (0.152)	0.084 (0.206)
Trading income/Employees	0.082*** (0.014)	0.070*** (0.013)	0.091 (0.058)	0.420** (0.187)	0.333** (0.104)
<i>B. Standard errors clustered only by labor market</i>					
Lending income/Employees	0.078*** (0.010)	0.048*** (0.006)	0.103* (0.057)	0.485*** (0.101)	0.084 (0.120)
Trading income/Employees	0.082*** (0.009)	0.070*** (0.009)	0.091** (0.043)	0.420*** (0.115)	0.333*** (0.101)
Employee FEs	Yes	Yes	Yes	Yes	Yes
Division $\times$ Hierarchy $\times$ Age $\times$ Tenure $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
No. of observations	122,486	51,291	58,460	4,102	3,457
No. of banks	12	12	11	11	7
Adjusted R^2	.90	.79	.89	.89	.79

Table 7: Table 6.

Table 7
Risk sharing outside the crisis after loan loss realizations

Dependent variable:	Bonus share				
	All D1-D8 (1)	Internal services D1+D2 (2)	Loan business D3+D4 (3)	Trading business D7+D8 (4)	Wealth mgmt. D5+D6 (5)
ΔLLP /Employees	-0.082*** (0.012)	-0.058*** (0.008)	-0.105*** (0.033)	-0.416** (0.197)	-0.307*** (0.157)
Employee FEs	Yes	Yes	Yes	Yes	Yes
Division $\times$ Hierarchy $\times$ Age $\times$ Tenure $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
No. of observations	124,632	52,528	59,240	4,102	3,467
No. of banks	13	13	12	11	8
Adjusted R^2	.90	.79	.89	.89	.79

Table 8: Table 7.

Read:

- Table 5 shows that both lending income and trading income raise bonus shares across the bank.
- In the all-division specification, an extra EUR 1 million in lending income per employee raises standardized bonuses by 2.6 percentage points, while an extra EUR 1 million in trading income per employee raises them by 2.0 percentage points.
- Cross-division spillovers are visible: internal service employees' bonuses respond positively to both lending and trading income.
- Table 6 repeats the analysis in the 2003–2007 employee panel with employee fixed effects and labor-market-by-year controls; the positive sensitivity remains.
- Table 7 shows that loan loss provisions lower bonus shares, and the effect appears across divisions, including internal services and wealth management.

Economic message: These tables are the strongest evidence against a pure individual-incentive story. A support employee's bonus responding to loan losses or trading income is not a natural individual-effort incentive, but it is exactly what a bankwide bonus pool predicts.

Interpretation: The employee fixed effects in Table 6 are especially important. They mean the result is not just that more productive or higher-paid employees work in banks with better lending or trading performance. The same employee's bonus varies with bankwide income over time.

5.6 Figure 4 and Figure 5: employee groups and labor-market comparison

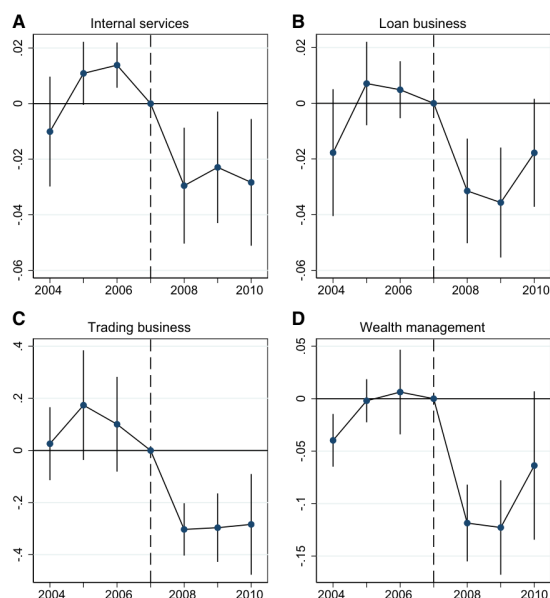


Figure 4: Figure 4.

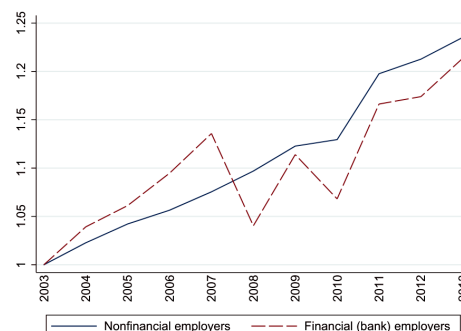


Figure 5: Figure 5.

Read:

- Figure 4 plots crisis-year bonus-share effects for internal services, loan business, trading business, and wealth management.
- Internal service employees also experience bonus-share declines in the crisis, despite limited responsibility for trading losses or loan portfolio deterioration.
- Trading employees experience large swings, consistent with both incentive and risk-sharing channels.
- Figure 5 compares wage growth of service-function employees inside and outside finance.
- Service wages outside the financial sector do not show the same contraction, which weakens the interpretation that the bank service-worker cuts are driven by a general labor-market shock.

Economic message: The two figures jointly separate bank-level risk sharing from broader labor-market wage adjustment. If accountants and IT workers in banks were merely reacting to outside options, similar nonfinancial workers should show comparable wage declines.

Interpretation: Figure 4 also speaks to fairness and governance. Employees who did not create the losses still absorbed part of them. This is efficient risk sharing from the shareholders' perspective, but it also means bankwide bonus pools expose all employees to aggregate bank risk.

5.7 Table 8 and Table 9: robustness on productivity and fixed effects

Table 8

Pre-crisis compensation differences across banks

Dependent variable:	log w_{2007}					
	(1)	(2)	(3)	(4)	(5)	(6)
Dummy: Δ ST debt funding 2008 > P50	0.008 (0.040)	-0.007 (0.037)				
Dummy: Δ off-BS activity 2008 > P50			-0.013 (0.041)	-0.051 (0.043)		
Dummy: Δ cash 2008 > P50					-0.059 (0.042)	-0.038 (0.043)
Bank characteristics in 2007:						
Loans / Assets		0.022 (0.096)		-0.038 (0.107)		0.012 (0.102)
Deposits / Assets		0.100 (0.123)		0.235 (0.156)		0.133 (0.124)
Equity / Assets		1.399** (0.568)		1.670* (0.986)		1.343** (0.623)
log assets		-0.017* (0.009)		-0.014 (0.011)		-0.015 (0.010)
Division $\times$ Hierarchy $\times$ Age $\times$ Tenure FEs	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	6,127	6,127	5,639	5,639	6,127	6,127
No. of banks	44	44	40	40	44	44
Adjusted R^2	.83	.84	.84	.85	.83	.84

Table 9: Table 8.

Table 9

Pay variation explained by employee fixed effects

Dependent variable:	$\Delta w_{it} = (w_{it} - w_{it-1}) / w_{it-1}$					
	Comparison 1		Comparison 2		Comparison 3	
	(1)	(2)	(3)	(4)	(5)	(6)
R^2	.15	.46	.24	.56	.29	.59
Adjusted R^2	.14	.09	.23	.23	.28	.29
Bank FEs	Yes	No	Yes	No	No	No
Employee FEs	No	Yes	No	Yes	No	Yes
Division $\times$ Hierarchy $\times$ Age $\times$ Tenure FEs	Yes	Yes	No	No	No	No
Division $\times$ Hierarchy $\times$ Age $\times$ Tenure $\times$ Year FEs	No	No	Yes	Yes	Yes	Yes
Bank $\times$ Year FEs	No	No	No	No	Yes	Yes
No. of observations	103,738	103,738	103,310	103,310	103,310	103,310
No. of banks	31	31	31	31	31	31

We report the R^2 and the adjusted R^2 of panel regressions in the subsample of employee-year observations with unique employee identifiers over the period 2003 to 2007. The dependent variable is the change of employees' annual total compensation $\Delta w_{it} = (w_{it} - w_{it-1}) / w_{it-1}$ and regressed either on bank or on employee fixed effects. Columns 1 and 2 include division $\times$ hierarchy $\times$ age $\times$ tenure fixed effects, whereas columns 3 to 6 include division $\times$ hierarchy $\times$ age $\times$ tenure $\times$ year fixed effects. Columns 5 and 6 additionally control for bank $\times$ year fixed effects.

Table 10: Table 9.

Read:

- Table 8 tests whether employees at banks that become more constrained in 2008 were already paid differently in 2007.
- The constraint dummies are small and statistically insignificant, suggesting no large precrisis compensation differences across constrained and unconstrained banks within labor-market cells.
- Table 9 compares how much pay variation is explained by bank/labor-market fixed effects versus employee fixed effects.
- Employee fixed effects increase the simple R^2 , but adjusted R^2 values are much closer, indicating that the rich labor-market controls explain meaningful variation without overfitting as severely as employee fixed effects.

Economic message: These robustness tables address the main alternative explanation: perhaps constrained banks had systematically different employees. The evidence suggests that precrisis pay and observable labor-market cells are broadly comparable.

Interpretation: The robustness results do not eliminate every sorting concern, but they make the interpretation more credible. The paper's main claims do not rest on a single regression; they rest on multiple tests that point to bankwide risk sharing.

6 Short summary

- The paper studies banker bonuses using granular payroll data from Austrian, German, and Swiss banks.
- The central thesis is that bonuses are not only incentive pay. They also share risk between shareholders and employees.
- Financial frictions make shareholder value concave in internal cash, so shareholders value wage flexibility in bad states.
- In the model, individual pay equals an incentive component times a bankwide wage scale.
- The bankwide wage scale is the risk-sharing component.
- Bonuses are widespread across the bank, even in support functions with limited direct profit attribution.
- Table 3 shows that risk sharing explains a nontrivial share of pay variation, especially for junior and nontrading employees.
- During the financial crisis, employees with higher precrisis bonus shares experienced larger pay cuts.
- The pay cuts are stronger in banks facing tighter financial constraints.
- Bonus shares respond to lending income, trading income, and loan loss provisions across divisions.
- These cross-divisional sensitivities are difficult to explain purely as individual incentives.
- The employee-panel tests with employee fixed effects support the cross-divisional risk-sharing interpretation.
- Service employees outside finance do not exhibit the same wage contraction, reducing concerns that the findings are driven only by labor-market outside options.
- The regulatory implication is nuanced: bonus caps may reduce incentive problems but also make labor costs more rigid.

7 Conclusion

7.1 Core conclusion

Efing, Hau, Kampkötter, and Rochet provide a complementary interpretation of bank bonus pay. Bonuses are not merely a way to motivate effort or reward individual performance. They also operate as a risk-sharing contract. When banks are financially constrained, variable pay allows the wage bill to fall, preserving internal cash and reducing the need for costly external finance.

7.2 What the paper changes about compensation interpretation

The paper changes how one should read a bank bonus. In the standard narrative, bonuses are tied to incentives and therefore potentially to excessive risk-taking. In this paper, part of the bonus is closer to a state-contingent wage claim. It lets the bank shift some aggregate downside risk to employees. This is why bonuses can be observed even in internal service divisions where individual performance is hard to map to bank profits.

7.3 Why the employee-level evidence matters

Most compensation research focuses on executives. This paper shows that nonexecutive employee pay is central for understanding bank cost flexibility because nonexecutives make up the overwhelming bulk of the wage bill. A bankwide bonus pool that affects thousands of employees can be a meaningful source of operating flexibility even if no single employee is a material risk taker.

7.4 Implications for financial stability

The financial-stability implication is not simply that bonuses are good or bad. Bonuses can encourage risk-taking when they reward upside without penalizing downside, but they can also help banks absorb negative shocks by lowering labor costs in distress. A regulation that forces compensation into fixed salary may reduce incentive distortions while increasing operating leverage. The optimal regulatory approach therefore depends on whether the policy targets incentive asymmetry, risk sharing, or both.

7.5 Implications for bonus regulation

A crude cap on bonuses may miss the distinction between bonus pay that creates risk-taking incentives and bonus pay that provides aggregate risk sharing. A more refined policy could focus on the design of bonus pay: deferral, clawbacks, downside participation, and bankwide performance conditions may preserve risk-sharing benefits while reducing short-term risk-taking incentives.

7.6 Implications for employees

From the employee perspective, bank bonuses expose workers to aggregate bank risk, including shocks they did not create. Service workers, junior workers, and employees far from trading desks can still lose bonus income when loan losses or trading losses occur elsewhere in the bank. The system can be efficient as risk sharing, but it also reallocates downside risk from shareholders to employees.

7.7 Implications for bank management

For managers, the paper implies that bonus-pool design is part of liquidity and capital management. Compensation structure can be used to reduce operating leverage before a crisis. A bank that relies heavily on fixed salary has less flexibility when funding dries up. A bank that uses variable compensation has a private stabilizer, but it must balance this against retention concerns and incentive distortions.

7.8 Research agenda

The paper opens several research directions. First, future work could examine whether risk-sharing bonuses reduce layoffs, capital raising, or fire sales during stress. Second, researchers could compare bank bonus systems across countries with different labor laws and bonus regulations. Third, the framework could be extended to deferred compensation and clawback regimes, which may provide risk sharing over longer horizons. Fourth, similar mechanisms may operate in other high-human-capital sectors such as consulting, law, private equity, and asset management.

7.9 Final takeaway

The main takeaway is that variable compensation is a financial contract, not just a managerial incentive tool. In banks, where labor costs are large and external finance is costly in bad states, bonus pay helps transform rigid labor costs into state-contingent claims. That flexibility can improve bank resilience, but it also means employees share risks that originate at the bankwide level.